Unit: 6 Unit 2: Exponents Radicals
Topic: Rationalizing Denominator

Name: ________________
Date: ____________

Quiz (Habanero)

Q1. What is the rationalized form of $\frac{1}{\sqrt{2}+\sqrt{3}}$?

(A) $\frac{\sqrt{2}-\sqrt{3}}{-1}$

(B) $\frac{\sqrt{3}-\sqrt{2}}{1}$

(C) $\frac{\sqrt{2}+\sqrt{3}}{2}$

(D) $\frac{\sqrt{6}-\sqrt{2}}{-1}$

(E) $\frac{\sqrt{6}+1}{-1}$

Q2. A weather monitoring station observes a temperature change of $-2\sqrt{3}$ degrees per hour. What is the rationalized form of $\frac{1}{\sqrt{3}+2}$ if the temperature changes $\frac{1}{\sqrt{3}+2}$ degrees?

(A) $\frac{\sqrt{3}-2}{-1}$

(B) $\frac{\sqrt{3}+2}{3}$

(C) $\frac{2-\sqrt{3}}{1}$

(D) $\frac{\sqrt{3}-2}{-1}$

(E) $\frac{2+\sqrt{3}}{1}$

Q3. A rainfall gauge measures $\frac{1}{\sqrt{2}}$ inches of rain in an ecosystem. What is the rationalized form of $\frac{1}{\sqrt{2}}$?

(A) $\frac{\sqrt{2}}{2}$

(B) $\frac{1}{2}$

(C) $\sqrt{2}$

(D) $\frac{1}{\sqrt{2}}$

(E) $\frac{2}{\sqrt{2}}$

Q4. What is the rationalized form of $\frac{2}{\sqrt{5}-1}$?

(A) $\frac{2(\sqrt{5}+1)}{4}$

(B) $\frac{\sqrt{5}-1}{2}$

(C) $\frac{\sqrt{5}+1}{2}$

(D) $\frac{2(\sqrt{5}-1)}{-4}$

(E) $\frac{2(\sqrt{5}+1)}{2}$

Q5. The atmospheric pressure in a certain region can be modeled by $\frac{1}{\sqrt{x}+2}$, where x is the altitude. What is the rationalized form of this expression?

(A) $\frac{\sqrt{x}-2}{x-4}$

(B) $\frac{\sqrt{x}+2}{x+4}$

(C) $\frac{\sqrt{x}-2}{-4}$

(D) $\frac{\sqrt{x}-2}{x-4}$

(E) $\frac{2-\sqrt{x}}{-4}$

Q6. What is the rationalized form of $\frac{3}{\sqrt{6}+\sqrt{2}}$?

(A) $\frac{3(\sqrt{6}-\sqrt{2})}{4}$

(B) $\frac{3(\sqrt{6}+\sqrt{2})}{6}$

(C) $\frac{\sqrt{6}-\sqrt{2}}{3}$

(D) $\frac{\sqrt{6}+\sqrt{2}}{4}$

(E) $\frac{\sqrt{6}-\sqrt{2}}{2}$

Q7. What is the rationalized form of $\frac{2}{\sqrt{3}+1}$?

(A) $\frac{2(\sqrt{3}-1)}{2}$

(B) $\frac{\sqrt{3}+1}{2}$

(C) $\frac{2(\sqrt{3}+1)}{4}$

(D) $\frac{2(\sqrt{3}-1)}{2}$

(E) $\frac{\sqrt{3}-1}{2}$

Q8. What is the rationalized form of $\frac{1}{\sqrt{5}-\sqrt{2}}$?

(A) $\frac{\sqrt{5}+\sqrt{2}}{3}$

(B) $\frac{\sqrt{5}-\sqrt{2}}{1}$

(C) $\frac{\sqrt{5}+\sqrt{2}}{1}$

(D) $\frac{\sqrt{2}-\sqrt{5}}{3}$

(E) $\frac{\sqrt{5}+\sqrt{2}}{3}$

Open Ended Questions (FRQ)

Q9. The water level in a certain ecosystem can be modeled by the expression $\frac{1}{\sqrt{x}-1}$, where x is the amount of rainfall. Rationalize the denominator and simplify. Show all your work and give the final rationalized form of the expression.

Q10. The temperature change in a certain environment can be modeled by $\frac{2}{\sqrt{x}+2}$, where x is the number of hours. Rationalize the denominator and simplify. Show all your work and give the final rationalized form of the expression.

Chef's Tips

- Always check for extraneous solutions when rationalizing denominators.
- Make sure to multiply the numerator and denominator by the conjugate of the denominator.
- Simplify the expression as much as possible after rationalizing the denominator.